

Interactive Effects of Seasonal-Demand Characteristics on Manufacturing Systems

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Abstract

In this paper, we identify different characteristics of seasonal demands and analyze their impact on manufacturing systems. In particular, we consider a single-stage production facility that has flexibility to adjust its production rates or inventory target levels. Using Markov decision process formulation, we analyze a model that minimizes the total cost under infinite horizon discounted and average cost criteria. Through extensive numerical studies, we reveal several insights related to the individual and joint impact of seasonal demand characteristics that are of practical interests to manufacturers.

Keywords: seasonal demands; flexible production capacity; dynamic programming

1 Introduction

Managing production and inventory presents a significant challenge to the manufacturing firms when demands for their products demonstrate seasonality. Often a season corresponds to length of time period over which a certain level of demand is expected to be realized. When seasons change, demand patterns switch from one level to the other. Although there is a certain degree of repeatability, in reality the demand realizations across seasons are a function of several different factors. Seasons differ in terms of their time scale (relative length of a season) and the difference in level of demands in each season. In many cases, there is significant variability and uncertainty in the length of individual seasons. Sometimes, the sequence of different demand seasons might themselves have uncertainty and demands in some seasons may be more skewed than in other seasons. As a result, manufacturers use a combination of strategies to respond to these variations. Some manufacturers respond to seasonality in demands by adjusting production rates, while others respond by adjusting inventory levels. Both of these strategies have their own benefits and cost implications, and therefore many manufacturers use a combination of these two strategies. Adjusting production rates requires investments in production flexibility (machine and/or labor), which may be expensive in some environments. For instance, companies such as John Deere, Harley Davidson adjust their production capacity by operating additional shifts of production and adjusting the size of their workforce. In contrast, addressing seasonality in demands using

flexible inventory policies requires building up inventory before the peak demand season. Such policies lead to high inventory holding costs and/or the risk of inventory obsolescence.

For a given manufacturing setting, the optimal policy that control production rates and inventory levels depends significantly on the characteristics of the underlying seasonal demand conditions. Most of the existing literature focuses on a detailed analysis for a particular setting of seasonal demands. Our focus in this paper is to instead develop a modeling framework that captures a wide variety of seasonal demand settings. We model the seasonal demands as a Markov modulated Poisson process (MMPP), which could capture variety of seasonal patterns such as stochastically monotone, random or deterministic sequence of seasons by suitably adjusting its phase transition matrix.

Using Markov decision process (MDP) formulation, we develop a model to minimize production, holding and backordering costs. From literature, it is known that the structure of the optimal policy for our model is a phase-dependent threshold-level policy, that determines when to stop production or run production at a certain production rate. However, our framework also allows us to compare the various characteristics of seasonal demands in terms of their impact — characteristics that do have a dominating impact on performance vs. characteristics that do not. These insights could be valuable as manufacturers make investments in information and decision support systems to optimize their manufacturing strategies.

The paper is organized as follows. In Section 2, we provide further description on the various characteristics of seasonal demands and the underlying questions related to each issue. We also review the relevant literature to summarize the work done in addressing some key aspects of seasonal demands. In Section 3, we describe the manufacturing system, formulate the system dynamics as a MDP, and present an analytical expression of the optimal policy. In Section 4, we conduct numerical studies and discuss their inferences. Section 5 provides the concluding remarks.

2 Background and prior work

This section consists of two subsections. In Section 2.1, we provide additional details on the various characteristics of seasonal demands and list the underlying questions related to these characteristics. These issues motivate the modeling framework that we develop in subsequent sections. In Section 2.2, we review the key literature to summarize the existing work related to seasonal demands.

2.1 Characteristics of seasonal demands

We discuss the following five key characteristics of seasonal demands: (i) difference in demands across seasons, (ii) average duration of season, (iii) variability in season duration, (iv) random shifts in sequence of seasons, and (v) skewed seasonality.

2.1.1 Difference in average demand across seasons

Difference in average demand across seasons varies significantly among different markets. For instance, sales of computer systems are moderately higher in peak demand periods compared to low demand periods, whereas sales of toys and apparel in peak demand periods are more than double the sales in low demand periods. No doubt, when the difference in average demand across seasons is high, the flexibility needed in terms of production rates and/or inventory target levels is also high. In some cases, demands within a season might be much higher than the maximum production capacity possible in that season. Therefore, it is important to know how differences in average demand across seasons impact the performance of manufacturing systems.

2.1.2 Average duration of season

The duration of a season is measured in terms of time units, days, weeks or months. Although, most often seasonal demands refer to demand variations across the annual time scale (where demands show particular trends for periods ranging from 2 to 4 months), similar demand variations also occur across much shorter time scales that range in terms of 2 to 4 weeks. These ‘seasonal’ trends could be due to factors such as market conditions, sales initiatives at the end of the month, weather patterns, or other consumer preferences. The duration of a season has an important effect on manufacturing operations. When season durations are long, the effects of neighboring seasons would be limited mainly to the transition period, and it is possible that manufacturing decisions taken during a season do not influence performance in other seasons. When season durations are short, the effect of a season spills over into other seasons, and decisions in one season have strong impact on the performance in other seasons. We compare the impact of both short and long seasons on the performance of manufacturing systems and determine how optimal costs and optimal policies are affected by duration of seasons.

2.1.3 Variability in season duration

Season durations are not always deterministic, they may last longer or shorter than the average duration. For instance, a summer market campaign by John Deere might boost their demand for longer durations. On the other hand, customers may be hesitant in making investments in new equipments when the peak demand season is shorter than expected. Variability in the season duration adversely affects the production and inventory policies. We investigate how variability in the season duration impacts the optimal costs and policies of manufacturing systems.

2.1.4 Random shifts in sequence of seasons

During the time horizon of interest, the sequence of seasons may not always be deterministic. If the sequence is deterministic, manufacturers have advance information to adjust production and inventory levels. However, due to economic effects or other weather factors, demands may skip certain seasons or switch randomly between seasons. In such cases, a manufacturer may have excess inventory, if demands skip the peak season. We investigate

the impact of random changes in sequence of seasons and compare their effects to situations where seasons follow a fixed sequence to understand the impact of this dimension of variability on performance.

2.1.5 Skewed seasonality

We use the term skewed seasonality to denote scenarios where both the average duration and the average demand during a season vary asymmetrically. This type of seasonal demand pattern is in contrast to balanced seasons, where the average duration is the same across the seasons (for instance 3 months per season) and the percentage difference in average demand between neighboring seasons is also equal (for instance 20% difference in demands across seasons). In particular, we focus on the following type of skewed seasonality which has practical interest: a seasonal demand process where the low demand season has a long duration, and the peak demand season has a short duration. Further, the demand in the peak season could be very high, and possibly even exceed the maximum production capacity. These types of seasonal demands are usually experienced by toy and apparel manufacturers, who sell a significant portion of their demand during the year-end holiday seasons. Skewed seasonal demands of this type is quite challenging for manufacturers; the peak demand season may be short enough to discourage investment in expensive capacity, but the demand in these seasons are also good revenue opportunities. We compare the impact of such skewness in seasonality and determine optimal production and inventory policies to deal with such variations.

2.2 Review of prior work

Past research have analyzed seasonal demands under a variety of settings, where the seasonal demands are modeled either as periodic demand process, time-varying demand process or Markov-modulated demand process. Karlin (1960) and Zipkin (1989) analyze inventory systems with periodic demands with no constraints on ordering. They demonstrate the optimality of a critical number policy that varies with respect to the demand periods. Kapuscinski and Tayur (1998) and Yossi and Federgruen (1997) analyze a manufacturing system with finite capacity and periodic demands (with period K) and determine the optimality of an order up to level policy for each of the K time periods. Both the papers (Yossi and Federgruen, 1997; Kapuscinski and Tayur, 1998) analyze systems with finite capacity, but they do not consider flexibility in capacity and its potential to reduce costs in systems with seasonal demands. Further, in their setting, the demand distributions are identical for every K -th period, while in our analysis, we consider both seasons with distinct demands and uncertainty in the sequence of seasons. Green and Kolesar (1991) model the average demand as a sinusoidal function of time. The average demand is characterized by two factors: amplitude and frequency; the amplitude denotes the maximum deviation of the demand from its average, and frequency is the rate at which the average demand completes one sinusoidal cycle. Through numerical studies, they demonstrate that the effects of non-stationary demands on the system can be ignored when the frequency is very low and the amplitude is not very high.

Graves (1999) analyze a system with non-stationary demands, which are modeled as an

integrated moving average process of order (0,1,1). They evaluate the performance of an adaptive base stock policy, where the base-stock level is adjusted as the demand forecast changes. Angelus and Porteus (2002) determine the optimal policy for a production system with short life products. For a finite horizon period where the demand distribution is not identical across time periods, they determine the optimal production capacity and production quantity for each time period. Iida (2002) analyze a non-stationary production-inventory model, where the cost parameters, production capacity and demands vary with time. The production capacity is stochastic and its distribution varies with time. Similarly, the demand is also stochastic and its distribution varies with time. Neale and Willems (2009) consider the problem of managing inventory at different nodes of a supply chain with non-stationary demands. They divide the planning horizon into different phases, such that the demand within each phase is stationary. Under the guaranteed service modeling framework, they determine the impact of non-stationary demands on the inventory control policy at each node of the supply chain. In contrast to these studies, we focus on analyzing individual and joint impact of seasonal demand characteristics on manufacturing systems by modeling seasonal demands as a MMPP.

There are several studies where seasonal demands are modeled as an MMPP (Song and Zipkin, 1993; Sethi and Cheng, 1997; Beyer and Sethi, 1997; Chen and Song, 2001; Kumar et al., 2013). Song and Zipkin (1993) determine optimal ordering policies for uncapacitated systems, whereas Beyer and Sethi (1997) and Sethi and Cheng (1997) analyze an inventory system using continuous state inventory variable, and show the optimality of a season-dependent policy for uncapacitated inventory systems. Chen and Song (2001) analyze a multi-stage inventory system, and show the optimality of a season-dependent echelon base-stock policy. Kumar et al. (2013) characterize the optimal policy for queueing systems with seasonal demands, where the service rate can be adjusted to minimize the total cost. They compare the performance of heuristic policies with that of optimal policy under different features of MMPP, which in our context relate to average difference of demand across seasons, average season duration and deterministic/random shifts in sequence of seasons. Gayon et al. (2009) analyze a differentiated pricing strategy in a manufacturing system with seasonal demands and finite capacity. They determine the optimal policy which simultaneously optimizes the pricing and target inventory level for each season. Thus, for systems where flexing capacity is expensive, the total average cost could be minimized by selectively admitting the customer through differentiated pricing. Mohebbi (2006) model a system that captures the uncertainty in both the production processes and the demand processes by modeling them using two independent Markov-modulated processes. In all of these studies, the MMPP model provides a degree of flexibility to analyze the impact of certain aspects of seasonality. However, most of them assume that the decision-maker has no control over the production rate. In our modeling efforts, we assess the importance of such flexibility in order to better respond to challenges due to seasonal demands.

Our work builds on the research presented in Bhat and Krishnamurthy (2015), which analyses single product manufacturing systems with seasonal demand conditions. They develop a model under MDP formulation, and establish the existence of optimal solution and policy under infinite horizon discounted and average cost criteria. Further, they derive structure of

the optimal policies under these criteria, and compare them with policies used in practice. In contrast, the objective of our paper is to understand the impact of individual and joint characteristics of different characteristics of seasonal demands on manufacturing systems. In the next section, we discuss our model which slightly generalizes the model in Bhat and Krishnamurthy (2015), and state results that are necessary for analytical completeness of the model.

3 The model

We consider a manufacturing system consisting of a production facility that replenishes an inventory buffer. Customer demands are satisfied from the inventory buffer if inventory is available, otherwise they are backordered. We assume that the production time is a random variable having Exponential distribution with mean $\frac{1}{\mu}$. We assume that the production rate μ could be adjusted to any value between 0 and $\mu_{max} < \infty$, albeit at different production cost rates.

We assume m demand seasons with distinct average demands in each of the m seasons. The seasonal demands are modeled as a MMPP with rm phases, such that the first r phases correspond to season 1, the next r phases correspond to season 2, and so on. We assume that each phase is completely observable to the manufacturer. This assumption is valid in the context of our analysis, where the upcoming seasons are either random due to the urgent component requirements by downstream assembly facilities or deterministic due to the demand forecast provided by the downstream assembly facilities, but the information on the new season is completely known at its onset. Since the duration of each phase of an MMPP is a random variable having Exponential distribution, the duration of each season is a sum of r Exponentially distributed random variables, which is a random variable having Erlang- r distribution. In this setting, the coefficient of variation of the season duration is $1/r$, where r can be adjusted to match the variability in the season duration. Demand arrivals in phase $s \in \{1, \dots, rm\}$ is a Poisson process with arrival rate $\lambda_s < \infty$. The demand arrival rates are equal in each of the r phases of a season. The phase transition process of the MMPP is governed by an external continuous time Markov chain with its generator matrix denoted as Q . The phases of the MMPP represent the state space of the Markov chain, i.e. $\mathbb{S} := \{1, 2, \dots, rm\}$. Note that the MMPP can also capture more generalized seasons, for instance, seasons differing in variability in duration can be modeled by suitably varying the number of Erlang phases across the seasons.

We formulate the system dynamics as a discrete epoch MDP using the uniformization technique, (Lippman, 1975). The decision epochs correspond to instances of demand arrival, production completion and phase change. The time interval between successive decision epochs is a random variable having Exponential distribution and mean $\frac{1}{\nu}$, where ν is the sum of the maximum rates of demand arrival, production and phase transition. The state of the MDP evolves as a two-dimensional Markov chain: $X(t) = \{(n, s) | n \in \mathbb{Z}, s \in \mathbb{S}\}$ where n is the net inventory of the system, s is the demand phase and $\mathbb{Z}$ is the set of integers. The net inventory is defined as the difference of the produced goods in the buffer and the

backorders.

We assume that the maximum production rate is greater than the phase-averaged arrival rate, i.e.

$$\mu_{max} > \sum_{s \in \mathbb{S}} \hat{\pi}_s \lambda_s \quad (1)$$

where $\{\hat{\pi}_1, \dots, \hat{\pi}_{rm}\}$ denote the stationary probability distribution over the rm phases. The production cost is assumed to be piecewise linear function of the production rate μ , with l linear segments. The production cost is formulated as follows:

$$c(\mu) = \begin{cases} c_1 \mu & 0 \leq \mu \leq \mu_1 \\ \vdots & \vdots \\ \sum_{i=1}^{l-1} c_i(\mu_i - \mu_{i-1}) + c_l(\mu - \mu_{l-1}) & \mu_{l-1} < \mu \leq \mu_l \end{cases} \quad (2)$$

where $0 = \mu_0 < \mu_1 < \dots < \mu_{l-1} < \mu_l = \mu_{max}$, and $0 < c_1 < c_2 < \dots < c_{l-1} < c_l < \infty$.

Besides production cost, the manufacturer also incurs inventory holding cost and backorder cost. The inventory holding cost per unit time per unit of inventory is h , and the backorder cost per unit time per unit of backorder is b . The discounted cost function for k -th transition, $C_{\alpha,k}(n, s, \mu)$, is given by:

$$\begin{aligned} C_{\alpha,k}(n, s, \mu) &= \int_0^\infty \left(\int_0^\tau C_k(n, s, \mu) e^{-\alpha t} dt \right) \nu e^{-\nu \tau} d\tau \\ &= \frac{1}{\alpha + \nu} C_k(n, s, \mu) \end{aligned}$$

where $C_k(n, s, \mu)$ is the cost function per unit time for the k -th transition such that $C_k(n, s, \mu) = c(\mu) + h(n)^+ + b(n)^-$. Without loss of generality, we rescale the time to set $\alpha + \nu = 1$.

The optimality equation under infinite horizon discounted expected cost criteria is given as follows:

$$\begin{aligned} v_\alpha(n, s) &= L v_\alpha(n, s) \\ &= \inf_{\mu \in \mathbb{A}} \left[C_\alpha(n, s, \mu) + \mu v_\alpha(n+1, s) + \lambda_s v_\alpha(n-1, s) + \sum_{s' \in \mathbb{S}, s' \neq s} Q_{ss'} v_\alpha(n, s') \right. \\ &\quad \left. + (\nu - \lambda_s - \mu + Q_{ss}) v_\alpha(n, s) \right] \quad \text{for } n \in \mathbb{Z} \text{ and } s \in \mathbb{S} \end{aligned} \quad (3)$$

where $v_\alpha(n, s)$ is the infinite horizon optimal discounted expected cost function, and $C_\alpha(n, s, \mu)$ is the discounted cost per transition, and is independent of the subscript k .

We next formulate the optimality equation under the infinite horizon average expected cost criterion. Let g denote the optimal average cost under the infinite horizon average expected

cost criterion, and $w(n, s)$ be the corresponding relative value function. Then,

$$g = \lim_{\alpha \downarrow 0} \alpha v_\alpha(n, s) \quad (4)$$

$$w(n, s) = \lim_{j \rightarrow \infty} (v_{\alpha_j}(n, s) - v_{\alpha_j}(n_0, s_0)) \quad (5)$$

$$\begin{aligned} w(n, s) + g \geq & \frac{1}{\nu} \min_{\mu \in \mathbb{A}} [C(n, s, \mu) + \mu w(n+1, s) + \lambda_s w(n-1, s) + \sum_{s' \in \mathbb{S}} Q_{ss'} w(n, s') \\ & + (\nu - \lambda_s - \mu) w(n, s)] \quad \text{for } n \in \mathbb{Z} \text{ and for } s \in \mathbb{S} \end{aligned} \quad (6)$$

where (n_0, s_0) is some distinguished state, and α_j is a converging sequence such that $\lim_{j \rightarrow \infty} \alpha_j = 0$. $C(n, s, \mu)$ is the cost per transition, and is independent of the subscript k .

Before discussing the structure of the optimal policy, we address the technical issues involved in formulating the optimality Equations 3 and 6. Since $C_\alpha(n, s, \mu)$ is unbounded, the operator L is not a single-stage contraction mapping, and the standard approach of using Banach fixed point theorem to determine the existence of the optimal solution is not applicable. However, from Theorem 11.5.3, Puterman (1994), it follows that the mapping L is a $J(>1)$ -stage contraction mapping, and the optimal solution and the stationary optimal policy exist under the infinite horizon discounted expected cost criterion. Similarly, the optimal solution to average cost optimality Equation 6 may not exist, since $C(n, s, \mu)$ is unbounded. However, the existence of the optimal solution follows from Theorem 1 of Sennott (1989). The assumptions of both these theorems hold for our model, and we refer the reader to Bhat and Krishnamurthy (2015) for validation of the assumptions.

We next discuss the structure of the optimal policy for our model under the infinite horizon discounted and average expected cost criteria.

Theorem 3.1 (Theorem 1, Bhat and Krishnamurthy (2015)). *The infinite horizon discounted cost optimal Equation 3 has a structured optimal policy characterized by finite inventory levels $z_{i,s}, i \in \{1, \dots, l\}$ for each phase $s \in \mathbb{S}$. The optimal production rate $\mu^*(n, s) = \mu_l$ for $n < z_{l,s}$, $\mu^*(n, s) = \mu_i$ for $z_{i+1,s} \leq n < z_{i,s}$ for $i \in \{1, \dots, l-1\}$, and $\mu^*(n, s) = 0$ for $n \geq z_{1,s}$.*

The proof of the above theorem is based on minimizing the μ -dependent terms from right side of Equation 3. Subsequently, by applying the first order minimization condition, the expression for the optimal policy is obtained. From Bhat and Krishnamurthy (2015), it also follows that the optimal policy under the infinite horizon discounted expected cost criterion converges to the optimal policy under the average expected cost criterion as the discount factor α approaches 0. Thus, the structure of the optimal policy is identical under both criteria.

Some literatures (e.g. Grubbström (1980); Van der Laan and Teunter (2002)) consider opportunity costs due to inventory investment in addition to the inventory costs considered in this paper. In such cases, the inventory holding parameter h needs to be suitably adjusted for the discounted cost optimal policy to converge to the average cost optimal policy.

Note that the structure of the production cost function strongly influences the structure of the optimal policy. The optimal policy consists of only l optimal production rates and l threshold levels for each phase since the production cost function consists of l linear segments (Equation 2). From the structure of the optimal policy, we also infer that the threshold levels $(z_{1,s}, \dots, z_{l,s})$ are adjusted r times in each season corresponding to each of the r phases. Note that as r increases, the variability in the season duration decreases (coefficient of variation of season duration: $\frac{1}{r}$). Adjusting threshold levels multiple times in seasons with lower variability in duration is often observed in practice. With reduction in the variability of the season duration, the decision-maker would have a better perspective of the season, and could leverage this information by adjusting the threshold levels multiple times in the season, for instance at the start, middle and end of the season.

In the following section, we analyze the impact of different characteristics of seasonal demands on the optimal costs and the optimal policies of manufacturing systems.

4 Impact of seasonal demands on manufacturing systems

Recall that we identified five key characteristics of seasonal demands: (i) difference in average demand across seasons, (ii) average duration of seasons, (iii) variability in season duration, (iv) random shifts in sequence of seasons, and (v) skewed seasonality. Through numerical studies, we investigate the individual and joint impact of these characteristics on manufacturing systems. We conduct the numerical studies under the infinite horizon average cost criterion, because the decisions are made so frequently that the discount factor over one decision epoch, $\frac{\nu}{\nu+\alpha}$ is close to 1. Also, the numerical analysis under the average expected cost criterion is robust, since unlike the discounted expected cost, the average expected cost is independent of the initial distribution of the state space. In order to simplify the analysis and keep the focus on the effects of seasonality, we consider a manufacturing system with production cost function that has two linear segments, $l = 2$, i.e.,

$$c(\mu) = \begin{cases} c_{reg}\mu & 0 \leq \mu \leq \mu_{reg} \\ c_{reg}\mu_{reg} + c_{max}(\mu - \mu_{reg}) & \mu_{reg} < \mu \leq \mu_{max} \end{cases} \quad (7)$$

where $0 < c_{reg} < c_{max}$ are scalar constants. The subscripts *reg* and *max* indicate the production mode status; *reg* indicates that the facility is producing at primary mode, where the maximum production rate is μ_{reg} ; *max* indicates that the facility is producing at secondary mode, where the maximum production rate is μ_{max} . We denote the corresponding optimal threshold levels $z_{1,s}$ and $z_{2,s}$ as $z_{b,s}$ and $z_{\mu,s}$ respectively.

We seek to determine how do the factors of seasonality influence the optimal average cost g and the optimal threshold levels $[z_{\mu,s}, z_{b,s}]$. We first describe the parameter settings, before we discuss the findings from the numerical study.

We consider a manufacturing system where μ_{reg} is set to 10 units/day and μ_{max} is set to 15 units/day. We assume that the manufacturer experiences three demand seasons ($m = 3$).

To determine the impact of different seasonal factors, we vary their parameters as follows:

4.1 Difference in average demand across seasons

We consider two cases, such that the difference in average demand across seasons vary by 2 and 4 units/day. Let $[\lambda_1, \lambda_2, \lambda_3]$ denote the average demands in seasons 1, 2 and 3, respectively. We set $[\lambda_1, \lambda_2, \lambda_3]$ as $[7, 9, 11]$, and $[5, 9, 13]$ units/day respectively for the two cases. Note that in both cases the season averaged demand is 9 units/day, which is 90% of regular production rate ($\mu_{reg} = 10$ units/day), but the average demand varies from 7 to 11 units/day across the seasons in the first case, and from 5 to 13 units/day across the seasons in the second case. Thus, analysis using these two cases will reveal the impact of difference in average demand across seasons, especially when compared with systems having identical average demand across the three seasons, i.e. $[9, 9, 9]$ units/day.

4.2 Average duration of seasons

The average season durations are equal across the three seasons. For cases other than skewed seasonality case, we set the average season duration to two values: 30 days and 100 days, that correspond to month-long season and quarter-of-a-year-long season, respectively. These two cases help us understand the impact of seasons with shorter and longer durations on the optimal cost and policy of the manufacturing system. The time horizon, i.e. sum of the average duration of the three seasons, for these two cases are 90 days and 300 days, respectively.

4.3 Variability in season duration

Each season consists of r phases of the Erlang- r probability distribution. We consider two cases $r = 1$ and $r = 3$. The first case $r = 1$ corresponds to Exponential distribution of the season duration, which has coefficient of variation 1; the second case $r = 3$ corresponds to Erlang-3 distribution of the season duration, which has co-efficient of variation 0.3. Thus, the first case models seasons with high variability in season duration, and the second case models seasons with low variability in its duration.

4.4 Random shifts in sequence of seasons

We consider two shifts in sequences of seasons: the deterministic sequence and the random sequence. In the deterministic sequence the season transitions are 1-2-3-1-2-3, and so on. In the random sequence, each season transits to one of the other two seasons with a probability of 0.5. We assume that for seasons with more than one phase ($r > 1$), the transition to the next season occurs only at the end of the r -th phase of the current season.

4.5 Skewed seasonality

For the skewed seasonality case, we set the ratio of average duration of the seasons, season 1 : season 2 : season 3 be equal to 3 : 2 : 1. Thus, the duration of low season (season 1) is thrice that of peak season (season 3), and the duration of moderate season (season 2) is twice that of peak season (season 3). We consider two sub-cases in skewed seasonality such that the time horizon in these cases are 90 days and 300 days, respectively. Note that these sub-cases are in line with our discussion related to shorter and longer average season durations. For the case with time horizon of 90 days, the durations of seasons 1, 2 and 3 are 45 days, 30 days and 15 days, respectively; for the case with time horizon of 300 days, the durations of seasons 1, 2 and 3 are 150 days, 100 days and 50 days, respectively. We set the average demand for the three seasons $[\lambda_1, \lambda_2, \lambda_3]$ as $[7, 9, 15]$ units/day, respectively. This ensures that the season averaged demand is 9 units/day. Also, note that the average demand in the peak season is more than twice that of the low-demand season and is equal to the maximum production rate μ_{max} .

The phase transition matrix Q is set such that for the case of skewed seasonal demands, the proportion of season duration (season 1 : season 2 : season 3) is (3 : 2 : 1). When the demands do not have skewed seasonality, the proportion of season durations (season 1 : season 2 : season 3) is (1 : 1 : 1).

Table 1 summarizes the different parameters of seasonal demand factors that are discussed above.

Table 1: Parameters of seasonal factors.

Seasonal factors	Parameters
Difference in average demand across seasons (units/day)	0, 2, 4
Time horizon (days)	90, 300
Average demand in skewed seasonality $[\lambda_1, \lambda_2, \lambda_3]$ (units/day)	$[7, 9, 15]$
Ratio of season duration in skewed seasonality [season 1: season 2: season 3]	$[3 : 2 : 1]$
Probability distribution of season duration	Exponential, Erlang-3
Shifts in sequence of seasons	Deterministic, Random

4.6 Impact of workload

Note that we set the season averaged demand (9 units/day) to 90% of regular production rate (10 units/day), but it is only 60% of maximum production rate (15 units/day). To capture the effects of seasonality at high workload levels, we also run the experiments with the above set of seasonal parameters at a season averaged demand of 11 units/day, which is 110% of regular production rate, and 73.33% of maximum production rate. For the two cases, where the difference in averaged demand across seasons varies by 2 and 4 units/day,

we set $[\lambda_1, \lambda_2, \lambda_3]$ to $[9, 11, 13]$, and $[7, 11, 15]$ units/day, respectively. For the skewed seasonality case, we set the average demand as $[9, 11, 17]$ units per day. Rest of the parameters remain the same as before.

The holding cost h equals \$1 per unit per day and the backorder cost b equals \$50 per unit per day. The high ratio $\frac{b}{h} = 50$ ensures that the average service level (percentage of the time the arriving customer demands are not backordered) is 98% under the optimal policy. The production costs are $c_{reg} = \$20$ per unit and $c_{max} = \$30$ per unit. The production cost parameters imply that the unit production cost increases by 50% when the average production rate increases by 50%, which is realistic in manufacturing settings. The optimal cost and the optimal threshold levels of the policies are estimated using policy iteration algorithm (Puterman (1994)) under the infinite horizon average cost criterion. The policy iteration algorithm is implemented using Matlab on a machine with Intel(R) Core(TM) i7-2600 CPU @ 3.40 GHz processor, installed memory of 8 GB, and 64-Bit Windows 7 operating system. For computational purposes, we truncate the state space such that the probability of reaching each of the boundary states is less than 10^{-8} under the optimal policy. Using this criteria, the state space is truncated to $[-1000:1000]$, i.e. from 1000 units of backorders through 1000 units of on-hand inventory. The average time required by the policy iteration algorithm is less than an hour and the maximum time over all cases did not exceed two hours.

We determine the impact of seasonal factors by comparing the resulting optimal cost and optimal policy to that of the system with stationary demands. Consider the stationary demand case where the average demand in each phase is 9 units/day. The optimal cost is then 5.876, and the optimal threshold levels $[z_{\mu,s}, z_{b,s}]$ in each phase are $[3, 13]$. This implies that, when the inventory level drops below 3 units, the manufacturer runs the production facility at maximum production rate (15 units/day), and when the inventory level is between 3 and 13, the manufacturer runs the production at regular production rate (10 units/day). The manufacturer stops the production when the inventory is 13 units or more. Let Δ denote the percentage increase in the optimal cost due to seasonality, i.e. if $Cost_{seasonal}$ denote the optimal cost under a certain seasonal demand case, and $Cost_{stationary}$ denote the optimal cost under stationary demand case, then $\Delta = \frac{Cost_{seasonal} - Cost_{stationary}}{Cost_{stationary}} \times 100$.

Table 2 lists the optimal cost and the optimal policy for 12 different cases of seasonal demands. Among these cases, we vary the sequence of seasons (deterministic (D) or random (R)), time horizon (90 or 300 days), difference in average demand across seasons (2 or 4 units/day), and also include the skewed seasonal demand case. For each of these 12 cases, the season duration is Exponentially distributed ($r = 1$), and the season averaged workload in 9 units/day.

4.7 Impact of difference of average demand across seasons and skewed seasonality

We first focus on the first three cases of Table 2. Cases 1 and 2 differ in terms of the average demand across the three seasons, while Case 3 considers the skewed seasonality case. The

Table 2: Exponentially distributed season durations with season averaged workload: 9 units/day.

Cases	Sequence (D/R)	Time horizon	Demand $[\lambda_1 \ \lambda_2 \ \lambda_3]$	Optimal cost g	Δ	Optimal Policy $\begin{bmatrix} z_{\mu,1} & z_{\mu,2} & z_{\mu,3} \\ z_{b,1} & z_{b,2} & z_{b,3} \end{bmatrix}$
1	D	90	$[7 \ 9 \ 11]$	5.963	1.47	$\begin{bmatrix} 1 & 3 & 7 \\ 7 & 15 & 22 \end{bmatrix}$
2	D	90	$[5 \ 9 \ 13]$	6.281	6.89	$\begin{bmatrix} 0 & 4 & 20 \\ 5 & 18 & 47 \end{bmatrix}$
3	D	90	$[7 \ 9 \ 15]$	6.750	14.87	$\begin{bmatrix} 1 & 25 & 51 \\ 9 & 39 & 81 \end{bmatrix}$
4	D	300	$[7 \ 9 \ 11]$	5.957	1.38	$\begin{bmatrix} 1 & 3 & 8 \\ 7 & 15 & 25 \end{bmatrix}$
5	D	300	$[5 \ 9 \ 13]$	6.259	6.51	$\begin{bmatrix} 0 & 3 & 23 \\ 5 & 15 & 53 \end{bmatrix}$
6	D	300	$[7 \ 9 \ 15]$	7.278	23.86	$\begin{bmatrix} 1 & 44 & 99 \\ 8 & 69 & 139 \end{bmatrix}$
7	R	90	$[7 \ 9 \ 11]$	5.972	1.62	$\begin{bmatrix} 1 & 3 & 7 \\ 7 & 11 & 22 \end{bmatrix}$
8	R	90	$[5 \ 9 \ 13]$	6.318	7.51	$\begin{bmatrix} 0 & 4 & 21 \\ 9 & 13 & 45 \end{bmatrix}$
9	R	90	$[7 \ 9 \ 15]$	6.996	19.05	$\begin{bmatrix} 4 & 15 & 55 \\ 27 & 37 & 92 \end{bmatrix}$
10	R	300	$[7 \ 9 \ 11]$	5.961	1.44	$\begin{bmatrix} 1 & 3 & 8 \\ 8 & 13 & 24 \end{bmatrix}$
11	R	300	$[5 \ 9 \ 13]$	6.272	6.73	$\begin{bmatrix} 0 & 3 & 22 \\ 5 & 15 & 56 \end{bmatrix}$
12	R	300	$[7 \ 9 \ 15]$	7.835	33.33	$\begin{bmatrix} 9 & 29 & 102 \\ 42 & 56 & 143 \end{bmatrix}$

sequence of seasons is deterministic, time horizon is 90 days, and average season duration is 30 days for the three cases. From the first two cases, we observe that as the difference of average demand increases from 2 units/day to 4 units/day, Δ increases from 2 to 7%. Δ increases further to 15% under the skewed seasonality case. *Thus, despite the season average workload being equal to 9 units per day in all the three cases, the optimal cost increases under seasonal demands due to imbalance in the demand distribution across the seasons. The imbalance in demand distribution increases the production, inventory and backorder cost especially for peak demand season. Higher imbalance in the demand distribution in the skewed seasonality case leads to further increase in the optimal cost, which we term as the severe effect of skewed seasonality.*

Next, we analyze the impact of these seasonal factors on the optimal threshold levels. Recall that for the stationary case, the optimal threshold levels $[z_{\mu,s}, z_{b,s}]$ in each of the three

seasons are $[3, 13]$. In Cases 1–3 of Table 2, we observe that the optimal threshold levels in season 1 (low demand season) is lower than $[3, 13]$, and the optimal thresholds levels in seasons 2 and 3 are higher than $[3, 13]$. The increase in the average demand in season 3 raises the threshold levels $[z_{\mu,s}, z_{b,s}]$ of season 3 in Cases 1–3. The threshold levels of season 2 increase due to the spillover effects of seasons.

4.8 Impact of seasons with longer average duration

We increase the time horizon from 90 days to 300 days (Cases 4–6) while retaining other parameters to determine the impact of seasonal conditions at longer season durations with average of 100 days. When compared to seasons with shorter durations (Cases 1–3), the optimal cost is slightly lower in Cases 4 and 5, and the optimal threshold levels are comparable between Cases 1 and 4, and Cases 2 and 5, respectively. The optimal cost reduces, since longer season duration limits the spillover effect of the peak demand in season 3 on seasons 1 and 2, thereby lowering the production cost in seasons 1 and 2. Further, the average demand in season 3 is less than the maximum production rate, hence longer duration of season 3 does not lead to significant increase in the optimal cost. In contrast, the optimal cost and the threshold levels of seasons 2 and 3 increase significantly for the skewed seasonality case (compare Cases 3 and 6). In the skewed seasonality case, the average demand in season 3 is as high as the maximum production rate, thereby leading to higher number of backorders. Longer duration of season 3 in such cases further increases the average number of backorders, leading to higher optimal cost and threshold levels. *Thus, we observe that increase in the season duration has contrasting effects under different seasonal demand conditions on manufacturing systems. The optimal cost and optimal threshold levels marginally decrease for seasonal demands where the peak demand is less than the maximum production rate, and increases significantly for the skewed seasonality case when the peak demand equals the maximum production rate.*

4.9 Impact of random shifts in sequence of seasons

We next investigate the impact of random shifts in sequence of seasons on manufacturing systems. Cases 7–12 in Table 2 have same parameters as in Cases 1–6, but with random shifts in sequence of seasons. When we compare Cases 7–12 (random sequence) with Cases 1–6 (deterministic sequence), we observe that the random shifts in sequence of seasons increase the optimal costs and threshold levels. However, the increase in the optimal cost is higher in cases with skewed seasonality (Cases 9 and 12) than in other cases (Cases 7, 8, 10, 11). The joint impact of random sequence of seasons and skewed seasonality has a similar impact on the optimal threshold levels across Cases 7–12. *Thus, the interaction between random shifts in the sequence of seasons and skewed seasonality need to be considered while assessing the impact of the seasonal factors on manufacturing systems.*

4.10 Impact of seasonal factors at higher workload levels

We next investigate the impact of seasonality at higher workload levels. To this end, we increase the average demand rate from 9 units/day to 11 units/day, and run the set of ex-

periments similar to Cases 1–12. Table 3 lists the corresponding 12 cases. As before, we compare the impact of seasonality by comparing the resulting optimal costs and optimal threshold levels with that of a system with identical season averaged demand (11 units/day) but having stationary demands. The optimal average cost for the stationary demand case with average demand 11 units/day is 7.490, and the optimal threshold levels $[z_{\mu,s}, z_{b,s}]$ are [8, 27].

Table 3: Exponentially distributed season durations with season averaged workload: 11 units/day.

Cases	Sequence (D/R)	Time horizon	Demand $[\lambda_1 \ \lambda_2 \ \lambda_3]$	Optimal cost g	Δ	Optimal Policy $\begin{bmatrix} z_{\mu,1} & z_{\mu,2} & z_{\mu,3} \\ z_{b,1} & z_{b,2} & z_{b,3} \end{bmatrix}$
13	D	90	[9 11 13]	7.700	2.81	$\begin{bmatrix} 3 & 10 & 19 \\ 15 & 32 & 47 \end{bmatrix}$
14	D	90	[7 11 15]	9.354	24.89	$\begin{bmatrix} 4 & 56 & 79 \\ 31 & 78 & 116 \end{bmatrix}$
15	D	90	[9 11 17]	10.668	42.44	$\begin{bmatrix} 28 & 109 & 152 \\ 54 & 141 & 188 \end{bmatrix}$
16	D	300	[9 11 13]	7.684	2.59	$\begin{bmatrix} 3 & 9 & 23 \\ 14 & 29 & 54 \end{bmatrix}$
17	D	300	[7 11 15]	10.587	41.35	$\begin{bmatrix} 2 & 93 & 143 \\ 28 & 130 & 185 \end{bmatrix}$
18	D	300	[9 11 17]	16.323	117.94	$\begin{bmatrix} 87 & 333 & 422 \\ 128 & 371 & 445 \end{bmatrix}$
19	R	90	[9 11 13]	7.701	2.82	$\begin{bmatrix} 4 & 9 & 21 \\ 17 & 29 & 51 \end{bmatrix}$
20	R	90	[7 11 15]	9.560	27.64	$\begin{bmatrix} 31 & 47 & 86 \\ 59 & 76 & 129 \end{bmatrix}$
21	R	90	[9 11 17]	11.418	52.45	$\begin{bmatrix} 81 & 107 & 166 \\ 115 & 142 & 196 \end{bmatrix}$
22	R	300	[9 11 13]	7.690	2.68	$\begin{bmatrix} 3 & 10 & 22 \\ 15 & 28 & 51 \end{bmatrix}$
23	R	300	[7 11 15]	11.094	48.12	$\begin{bmatrix} 57 & 82 & 152 \\ 93 & 116 & 195 \end{bmatrix}$
24	R	300	[9 11 17]	18.629	148.73	$\begin{bmatrix} 242 & 298 & 466 \\ 287 & 356 & 472 \end{bmatrix}$

To investigate the impact of difference in average demand across seasons and skewed seasonality at higher workload levels, we compare Cases 13–15 with Cases 1–3. We observe that Δ is significantly higher in Cases 13–15 compared to Cases 1–3. Although the workload difference between Case 3 and Case 15 is 22% (9 units/day vs. 11 units/day), the Δ corresponding to Case 3 is 15%, and for Case 15 it is 42%. Further, the optimal threshold levels also rise significantly, especially in peak demand seasons at higher workload levels,

leading to higher inventory levels and quicker shifts to running facility at higher production rate. We make similar observations for seasons with longer durations, when we compare Cases 16–18 with Cases 4–6. In particular, on comparing Case 5 with Case 17, where the difference of average demand across seasons is 4 units per day, Δ increases from 6 to 41%. Also comparing Case 6 with Case 18, i.e. the skewed seasonality cases, Δ increases from 24 to 118%. Similarly, the optimal threshold levels also surge in Cases 13–18 as compared to Cases 1–6. *We thus observe the amplifying effects of higher workload levels, where higher workload levels significantly amplify the impact of seasonality on the optimal cost and the optimal threshold levels of the manufacturing systems.*

In the next section, we investigate whether the impact of random shifts in sequence of seasons also amplifies at higher workload levels.

4.11 Impact of random shifts in sequence of seasons at higher workload levels

Cases 19–21 and 22–24 list the optimal cost and threshold levels for systems with random sequence of seasons and with time horizon 90 days and 300 days, respectively. We compare them with the optimal cost and optimal threshold levels of similar systems but at lower workload level of 9 units/day (Cases 7–12). We observe that the increase in the optimal cost due to the random sequence of seasons for the skewed seasonality cases is considerably higher at higher workload levels (Cases 21 and 24) than at lower workload levels (Cases 9 and 12). Further, the impact of random shifts in sequence of seasons are also significant for systems where the peak demand does not exceed the maximum production rate (compare Cases 14 and 20, Cases 17 and 23). Similar observations are made for the optimal threshold levels in seasons 2 and 3 of these cases. *Thus from these observations, we conclude that the impact of random shifts in sequence of seasons is amplified at higher workload levels on the optimal cost and threshold levels of the manufacturing systems.*

4.12 Impact of variability in durations of seasons

In this section, we determine the joint impact of variability in season duration and other factors of seasonality on the manufacturing system. We model the duration of each season as an Erlang-3 process. Since the optimal threshold levels $[z_{\mu,s}, z_{b,s}]$ are set for each phase, the manufacturer adjusts the production rate thrice in each season. Thus for three seasons, we specify the optimal threshold levels for 9 phases. Recall that in the Exponential distribution case, the manufacturer could adjust thresholds only once per season. Hence, we had specified the optimal threshold levels thrice, one for each season. For cases with Exponentially distributed seasons, we specified the averaged demand for each of the three phases. In the Erlang-3 distributed seasons, there are 9 phases with identical average demands in the three phases of each season. Hence for brevity in Table 4, we specify the average demand for the first phase of each season, i.e., phases 1, 4, and 7.

As before, we compare the impact of seasonality by comparing the resulting optimal costs and threshold levels with that of systems with stationary demands. For systems with stationary

demands and average demand of 9 units/day, the optimal cost is 5.876 and the optimal threshold levels $[z_{\mu,s}, z_{b,s}]$ for each of the 9 phases are [3, 13]. Table 4 lists the 12 cases that have similar parameter settings as in Table 2, but with Erlang-3 distributed season durations.

Table 4: Erlang-3 distributed season durations with season averaged workload: 9 units/day.

Case	Sequence	Time	Demand	Optimal cost	Δ	Optimal Policy																	
	(D/R)	horizon	$\begin{bmatrix} \lambda_1 & \lambda_4 & \lambda_7 \end{bmatrix}$	g		$\begin{bmatrix} z_{\mu,1} & z_{\mu,2} & z_{\mu,3} & z_{\mu,4} & z_{\mu,5} & z_{\mu,6} & z_{\mu,7} & z_{\mu,8} & z_{\mu,9} \\ z_{b,1} & z_{b,2} & z_{b,3} & z_{b,4} & z_{b,5} & z_{b,6} & z_{b,7} & z_{b,8} & z_{b,9} \end{bmatrix}$																	
25	D	90	$\begin{bmatrix} 7 & 9 & 11 \end{bmatrix}$	5.961	1.44		1	1	1	3	3	4	8	9	7								
26	D	90	$\begin{bmatrix} 5 & 9 & 13 \end{bmatrix}$	6.251	6.37		0	0	0	3	4	10	25	20	14								
27	D	90	$\begin{bmatrix} 7 & 9 & 15 \end{bmatrix}$	6.380	8.57		1	1	1	3	10	28	44	38	25								
28	D	300	$\begin{bmatrix} 7 & 9 & 11 \end{bmatrix}$	5.955	1.34		1	1	1	3	3	3	9	9	7								
29	D	300	$\begin{bmatrix} 5 & 9 & 13 \end{bmatrix}$	6.256	6.46		0	0	0	3	3	6	27	24	21								
30	D	300	$\begin{bmatrix} 7 & 9 & 15 \end{bmatrix}$	6.700	14.02		1	1	1	3	10	54	89	74	52								
31	R	90	$\begin{bmatrix} 7 & 9 & 11 \end{bmatrix}$	5.966	1.52		1	1	1	3	3	3	8	9	6								
32	R	90	$\begin{bmatrix} 5 & 9 & 13 \end{bmatrix}$	6.282	6.90		0	0	1	3	3	5	25	22	15								
33	R	90	$\begin{bmatrix} 7 & 9 & 15 \end{bmatrix}$	6.540	11.29		1	1	15	3	7	21	48	38	24								
34	R	300	$\begin{bmatrix} 7 & 9 & 11 \end{bmatrix}$	5.957	1.37		1	1	1	3	3	3	9	9	8								
35	R	300	$\begin{bmatrix} 5 & 9 & 13 \end{bmatrix}$	6.266	6.63		0	0	0	3	3	4	24	24	20								
36	R	300	$\begin{bmatrix} 7 & 9 & 15 \end{bmatrix}$	6.987	18.90		1	1	30	3	4	45	93	72	55								

From Cases 25–27, we observe that the impact of seasonal factors on the optimal cost and the optimal threshold levels is similar to that of systems with Exponentially distributed season durations (Cases 1–3). However, the increase in the optimal cost/threshold levels due to seasonal factors is marginally lower in Erlang-3 distributed season durations compared to Exponentially distributed season durations. Similar observations are made for seasons with longer durations (compare Cases 4–6 with Cases 28–30). In systems where seasons have Erlang-3 distributed durations, the manufacturer has additional flexibility to adjust the threshold levels thrice in each season. Moreover, the lower variability in Erlang-3 dis-

tributed season durations lowers the optimal cost compared to systems with Exponentially distributed season durations. *Thus, we observe the dampening effect of lower variability in season durations, which reduces the impact of seasonality on the optimal costs and threshold levels of the manufacturing systems.*

4.13 Impact of random shifts in sequence of seasons at lower variability in season duration

In this section, we analyze the impact of random shifts in sequence of seasons at lower variability in season duration. To this end, we compare the optimal costs as the shifts in the sequence of season change from deterministic (Cases 25–30) to random for systems with Erlang-3 distributed season durations (Cases 31–36) with that of systems having Exponentially distributed season durations with deterministic shifts in seasons (Cases 1–6) and random shifts in sequence of seasons (Cases 7–12), respectively. We observe that though the optimal cost of Cases 31–36 is higher than that of Cases 25–30 respectively, the increase in the optimal cost is lower compared to the cases with Exponentially distributed season durations. *Thus, we infer that the lower variability in season duration lowers the impact of random shifts in sequence of seasons on the manufacturing system.*

4.14 Impact of variability in season durations at higher workload levels

In Sections 4.10 and 4.11, we observed that the higher workload levels amplify the impact of seasonal factors on manufacturing systems. In this section, we investigate whether the higher workload levels also amplify the impact of variability in season durations on manufacturing systems. We run a similar set of 12 cases as listed in Table 4, but at a higher average demand of 11 units/day. The results are reported in Table 5. As before, we estimate the impact of these factors by comparing the resulting optimal costs and optimal threshold levels with that of systems with stationary demands. Recall that for systems with stationary demands and with average demand of 11 units/day, the optimal average cost is 7.490 and the threshold levels $[z_{\mu,s}, z_{b,s}]$ are $[8, 27]$.

Table 5 lists the 12 cases with Erlang-3 distributed season durations and with season averaged duration of 11 units/day. We first consider systems with deterministic sequence of seasons. We compare the increase in Δ due to seasonal factors at higher workload levels under Erlang-3 and Exponentially distributed season durations, i.e. we compare the increase in Δ from Cases 1–6 to Cases 13–18 with the increase in Δ from Cases 25–30 to Cases 37–42. We observe that similar to cases with Exponentially distributed seasons, Δ increases with the workload levels for systems with Erlang-3 distributed seasons but with significantly lower magnitude. *Thus, the dampening effects of low variability in season duration limit the surge in the optimal cost due to seasonal factors at higher workload levels.*

Table 5: Erlang-3 distributed season durations with season averaged workload: 11 units/day.

Case	Sequence	Time	Demand	Optimal cost	Δ	Optimal Policy													
	(D/R)	horizon	$\begin{bmatrix} \lambda_1 & \lambda_4 & \lambda_7 \end{bmatrix}$	g		$\begin{bmatrix} z_{\mu,1} & z_{\mu,2} & z_{\mu,3} & z_{\mu,4} & z_{\mu,5} & z_{\mu,6} & z_{\mu,7} & z_{\mu,8} & z_{\mu,9} \\ z_{b,1} & z_{b,2} & z_{b,3} & z_{b,4} & z_{b,5} & z_{b,6} & z_{b,7} & z_{b,8} & z_{b,9} \end{bmatrix}$													
37	D	90	$\begin{bmatrix} 9 & 11 & 13 \end{bmatrix}$	7.683	2.58		3	3	3	9	10	16	21	22	15				
38	D	90	$\begin{bmatrix} 7 & 11 & 15 \end{bmatrix}$	8.590	14.66		1	1	2	23	42	56	68	57	39				
39	D	90	$\begin{bmatrix} 9 & 11 & 17 \end{bmatrix}$	8.688	0.323		3	4	8	41	56	81	94	73	47				
40	D	300	$\begin{bmatrix} 9 & 11 & 13 \end{bmatrix}$	7.680	2.54		3	3	3	8	9	11	24	24	21				
41	D	300	$\begin{bmatrix} 7 & 11 & 15 \end{bmatrix}$	9.337	24.67		1	1	1	13	56	97	131	105	80				
42	D	300	$\begin{bmatrix} 9 & 11 & 17 \end{bmatrix}$	10.714	43.05		3	3	5	90	158	230	267	210	138				
43	R	90	$\begin{bmatrix} 9 & 11 & 13 \end{bmatrix}$	7.687	2.63		3	3	6	9	9	11	23	21	17				
44	R	90	$\begin{bmatrix} 7 & 11 & 15 \end{bmatrix}$	8.800	17.49		1	11	37	13	31	49	71	59	43				
45	R	90	$\begin{bmatrix} 9 & 11 & 17 \end{bmatrix}$	9.202	22.86		4	26	60	28	50	70	98	78	54				
46	R	300	$\begin{bmatrix} 9 & 11 & 13 \end{bmatrix}$	7.683	2.58		3	3	3	9	10	16	21	22	15				
47	R	300	$\begin{bmatrix} 7 & 11 & 15 \end{bmatrix}$	9.745	30.11		1	6	69	10	43	86	130	109	81				
48	R	300	$\begin{bmatrix} 9 & 11 & 17 \end{bmatrix}$	12.286	64.04		3	75	179	65	136	203	274	220	151				

4.15 Joint impact of random shifts in sequence of seasons and variability in season duration at higher workload levels

In Sections 4.10 and 4.11, we inferred that the impact of random sequence of seasons is higher at higher workload levels. Similar observations are made for systems with Erlang-3 distributed seasons. However, the increase in the optimal cost is significantly higher than for systems with Exponentially distributed seasons. For illustration, first consider the Exponential distribution case. At lower workload level, the optimal cost for systems with random sequence of seasons (Case 12) is 8% higher than that for systems with deterministic shifts in sequence of seasons (Case 6). However, at higher workload level, the optimal cost for systems with random sequence of seasons (Case 24) is 14% higher than that for systems with deterministic sequence (Case 18). Thus, for systems with Exponential distributed season durations, the increase in the optimal cost due to random sequence of seasons at higher workload levels is almost two times than that at lower workload levels. Now consider the Erlang-3 distributed seasons. At lower workload level, the optimal cost for systems with random sequence (Case 36) is 4% higher than that for systems with deterministic sequence of seasons (Case 30). However, at higher workload level, the optimal cost for systems with random sequence of seasons (Case 48) is 15% higher than that for systems with deterministic sequence of seasons (Case 42). Thus, the increase in the optimal cost due to random sequence of seasons at higher workload levels is almost four times than that at lower workload levels for systems with Erlang-3 distributed season durations. *Thus, we infer that the impact of random sequence of seasons increases at a higher rate with workload levels for systems with lower variability in duration of the seasons.*

We would also like to comment on our modeling assumptions related to season duration. The Exponentially distributed season duration has a coefficient of variation 1, where as the Erlang-3 distribution has a coefficient of variation 0.3. Despite the reduction in the coefficient of variation reduced from 1 to 0.3, the reduction in the optimal cost and the optimal threshold levels were not significant except in few cases, where season durations were longer with skewed seasonality conditions (Cases 18 and 42, Cases 24 and 48). However, note that the qualitative nature of the managerial insights remained the same irrespective of the co-efficient of variation of the season duration. Thus, we can safely assume that the qualitative nature of our managerial insights are valid even for systems with deterministic season durations, i.e. for season durations with Erlang- r distribution with a large value of r .

4.16 Practical application

The analytical models developed in this paper strike a balance between practical relevance and analytical complexity. Our choice of Markov-modulated Poisson process for demands and phase-type distributions allow us to derive structural results of optimal policies and analyze the impact of various seasonality settings observed in practice (difference in average demand across seasons, mean duration of seasons, variability in season duration, shifts in sequence of seasons and skewed seasonality). Thus, our models thus provide several insights to practitioners on the relative impact of such seasonality on optimal policies. Further, our modeling framework could be easily adapted through use of more elaborate phase-type dis-

tributions to more closely model the distributions of production times, inter-arrival time, and season duration observed in industry. Although, this will result in an increase in computational requirements, we do not expect that such details would lead to significant differences in the qualitative nature of the insights derived from our analysis. Also, by suitably adjusting the generator matrix Q of the Markov-modulated Poisson process, a wide variety of seasonal demand patterns and non-stationary demand patterns can be incorporated into our models. It is for this reason that such models of demand processes are very popular in the research literature. Finally, the structure of the optimal policies, makes it easy to incorporate the insights from our models into practice. We expect that insights derived from this paper can be easily incorporated in the form of decision rules within scheduling systems and software used in industry.

5 Concluding remarks

In this paper, we determine the impact of several seasonal factors on manufacturing systems. We model seasonal demands as a MMPP. In particular, we consider a manufacturing system consisting of a single-stage production facility with flexibility in adjusting the production rates and inventory target levels. Using MDP formulation, we develop a model that minimizes the total cost under the infinite horizon discounted expected cost and average expected cost criteria. The optimal policy for the model consists of season-dependent inventory threshold levels that determine whether to run the production at a certain production rate or to stop the production.

Through extensive numerical studies, we then investigate the impact of five key characteristics of seasonal demands: (i) difference in average demand across seasons, (ii) average season duration, (iii) variability in season duration, (iv) random sequence of seasons and (v) skewed seasonality. We determine the individual as well as joint impact of these characteristics at different workload levels on the optimal costs and the optimal threshold levels of manufacturing systems which are summarized below.

We observe severe effects of skewed seasonal demand conditions. Skewed seasonality significantly increases the optimal costs compared to other seasonal demand factors. The optimal cost reduces as season duration increases for cases where the peak demand is less than the maximum production rate; but for skewed seasonal demand conditions, the optimal cost increases with season duration. Also, the impact of random sequence of seasons is much higher for skewed seasonal conditions. We also observe that higher workload levels amplify the impact of seasonal factors and random sequence of seasons, whereas lower variability in season durations dampen them. The joint impact of high workload levels and lower variability in season duration has interesting effects. On the one hand, we observe that lower variability in season durations limits the surge in the optimal cost due to seasonal factors at higher workload levels. On the other hand, the impact of random sequence of seasons increases at a higher rate with workload levels for systems with lower variability in season durations.

These insights provide strong understanding into the production and inventory management issues faced by manufacturers under seasonal demand conditions. The framework developed in this paper is fairly general and helps to analyze the impact of several aspects of seasonality on manufacturing systems. This work can be extended to analyze systems with more general non-stationary demand patterns with less well-defined seasons. The work could also be extended to analyze systems with multiple products and/or multiple manufacturing stages.

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